CHAPTER - 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

The Nykaa E-Commerce Database Management System is designed to efficiently manage customers, categories, suppliers, products, inventory, orders, payments, and administrators. Functional dependencies help identify the relationship between attributes in each table, while redundancy analysis and data anomaly detection ensure data consistency and reduce duplication. A well-designed database improves performance, maintains data integrity, and supports efficient order management.

1. CUSTOMER TABLE

Attribute	Key Type
Customer_ID	Primary Key
First_Name	Not Null
Last_Name	Not Null
Gender	-
Date_of_Birth	-
Email	Unique Key
Phone	Unique Key
Password	-
Address	-
City	-
State	-
Pincode	-
Registration_Date	-

Functional Dependency

Customer_ID → First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Password, Address, City, State, Pincode, Registration_Date

Redundancy Analysis

Customer information is stored only once in the Customer table. Orders and Payments reference only Customer_ID, avoiding duplicate customer records.

Data Anomalies

- **Insert Anomaly:** A customer can register without placing an order.
- **Update Anomaly:** Updating customer details is done only in the Customer table.
- **Delete Anomaly:** Deleting an order does not delete customer information.

2. CATEGORY TABLE

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-
Status	-

Functional Dependency

Category_ID → Category_Name, Description, Status

Redundancy Analysis

Each category is stored only once. Products reference Category_ID instead of repeating category details.

Data Anomalies

- **Insert Anomaly:** Categories can be added before products.
- **Update Anomaly:** Category changes occur only once.
- **Delete Anomaly:** Deleting a product does not remove the category.

3. SUPPLIER TABLE

Attribute	Key Type
Supplier_ID	Primary Key
Supplier_Name	Not Null
Contact_Number	Unique Key
Email	Unique
Address	-

Functional Dependency

Supplier_ID → Supplier_Name, Contact_Number, Email, Address

Redundancy Analysis

Supplier details are stored only once. Products reference Supplier_ID.

Data Anomalies

- **Insert Anomaly:** Suppliers can be added before supplying products.
- **Update Anomaly:** Supplier information is updated only once.
- **Delete Anomaly:** Deleting a product does not remove supplier details.

4. PRODUCT TABLE

Attribute	Key Type
Product_ID	Primary Key
Category_ID	Foreign Key
Supplier_ID	Foreign Key
Product_Name	Not Null
Brand	-
Price	Check
Stock	Check
Ingredients	-
Skin_Type	-
Description	-
Expiry_Date	-

Functional Dependency

Product_ID → Category_ID, Supplier_ID, Product_Name, Brand, Price, Stock, Ingredients, Skin_Type, Description, Expiry_Date

Redundancy Analysis

Product details exist only in the Product table. Orders and Inventory reference Product_ID.

Data Anomalies

- **Insert Anomaly:** Products can be added before sales.
- **Update Anomaly:** Price or stock updates occur only once.
- **Delete Anomaly:** Deleting an order does not remove product information.

5. INVENTORY TABLE

Attribute	Key Type
Inventory_ID	Primary Key
Product_ID	Foreign Key
Quantity_Available	Check
Last_Updated	-

Functional Dependency

Inventory_ID → Product_ID, Quantity_Available, Last_Updated

Redundancy Analysis

Inventory information is stored separately, preventing duplication of stock details.

Data Anomalies

- **Insert Anomaly:** Inventory can exist before sales.
- **Update Anomaly:** Stock changes affect only Inventory.
- **Delete Anomaly:** Removing inventory does not delete products.

6. ORDERS TABLE

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Total_Amount	-
Order_Status	Check

Functional Dependency

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status

Redundancy Analysis

Orders store only transaction details while customer details remain in the Customer table.

Data Anomalies

- **Insert Anomaly:** Orders require an existing customer.
- **Update Anomaly:** Order status changes only in Orders.
- **Delete Anomaly:** Removing an order does not delete customer records.

7. ORDER_DETAILS TABLE

Attribute	Key Type
Order_Detail_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	Check
Price	Check

Functional Dependency

Order_Detail_ID → Order_ID, Product_ID, Quantity, Price

Redundancy Analysis

This table connects Orders and Products without repeating product information.

Data Anomalies

- **Insert Anomaly:** Products can be added only to existing orders.
- **Update Anomaly:** Quantity changes affect only Order Details.

- **Delete Anomaly:** Removing an order item does not delete the product.

8. PAYMENT TABLE

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	Check
Payment_Date	-
Amount	Check

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount

Redundancy Analysis

Payment information is stored separately from Orders, preventing duplicate payment records.

Data Anomalies

- **Insert Anomaly:** Payments are recorded after order creation.
- **Update Anomaly:** Payment status changes only in Payment.
- **Delete Anomaly:** Deleting payment records does not remove orders.

9. ADMIN TABLE

Attribute	Key Type
Admin_ID	Primary Key
Admin_Name	Not Null
Email	Unique
Password	-
Role	-

Functional Dependency

Admin_ID → Admin_Name, Email, Password, Role

Redundancy Analysis

Administrator information is stored once and referenced wherever administrative actions are required.

Data Anomalies

- **Insert Anomaly:** Admin accounts can be created independently.
- **Update Anomaly:** Admin details are updated only in the Admin table.

- **Delete Anomaly:** Deleting an admin account does not affect customer or order records.

10.CONCLUSION

The Nykaa E-Commerce Database System is designed using proper functional dependencies and normalized tables. Each entity stores only its relevant data, reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains integrity, and ensures efficient database management for the e-commerce system.